

National Coverage Determination (NCD)

Electrocardiographic Services

20.15

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Electrocardiographic Services

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Description Information

Benefit Category

Diagnostic Tests (other)

Please Note: This may not be an exhaustive list of all applicable Medicare benefit categories for this item or service.

Indications and Limitations of Coverage

Reimbursement may be made under Part B for electrocardiographic (EKG) services rendered by a physician or incident to his/her services or by an approved laboratory or an approved supplier of portable X-ray services. Since there is no coverage for EKG services of any type rendered on a screening basis or as part of a routine examination, the claim must indicate the signs and symptoms or other clinical reason necessitating the services.

A separate charge by an attending or consulting physician for EKG interpretation is allowed only when it is the normal practice to make such charge in addition to the regular office visit charge. No payment is made for EKG interpretations by individuals other than physicians.

On a claim involving EKG services furnished by a laboratory or a portable X-ray supplier, identify the physician ordering the service and, when the charge includes both the taking of the tracing and its interpretation, include the identity of the physician making the interpretation. No separate bill for the services of a physician is paid unless it is clear that he/she was the patient's attending physician or was acting as a consulting physician. The taking of an EKG in an emergency, i.e., when the patient is or may be experiencing what is commonly referred to as a heart attack, is covered as a laboratory service or a diagnostic service by a portable X-ray supplier only when the evidence shows that a physician was in attendance at the time the service was performed or immediately thereafter.

The documentation required in the various situations mentioned above must be furnished not only when the laboratory or portable X-ray supplier bills the patient or carrier for its service, but also when such a facility bills the attending physician who, in turn, bills the patient or carrier for the EKG services. (In addition to the evidence required to document the claim, the laboratory or portable X-ray supplier must maintain in its records the referring physician's written order and the identity of the employee taking the tracing.)

Long Term EKG Monitoring, also referred to as long-term EKG recording, Holter recording, or dynamic electrocardiography, is a diagnostic procedure which provides a continuous record of the electrocardiographic activity of a patient's heart while he is engaged in his daily activities.

The basic components of the long-term EKG monitoring systems are a sensing element, the design of which may provide either for the recording of electrocardiographic information on magnetic tape or for detecting significant variations in rate or rhythm as they occur, and a component for either graphically recording the electrocardiographic data or for visual or computer assisted analysis of the information recorded on magnetic tape. The long-term EKG permits the examination in the ambulant or potentially ambulant patient of as many as 70,000 heartbeats in a 12-hour recording while the standard EKG which is obtained in the recumbent position, yields information on only 50 to 60 cardiac cycles and provides only a limited data base on which diagnostic judgments may be made.

Many patients with cardiac arrhythmias are unaware of the presence of an irregularity in heart rhythm. Due to the transient nature of many arrhythmias and the short intervals in which the rhythm of the heart is observed by conventional standard EKG techniques, the offending arrhythmias can go undetected. With the extended examination provided by the long-term EKG, the physician is able not only to detect but also to classify various types of rhythm disturbances and waveform abnormalities and note the frequency of their

occurrence. The knowledge of the reaction of the heart to daily activities with respect to rhythm, rate, conduction disturbances, and changes are of great assistance in directing proper therapy and rehabilitation.

This modality is valuable in both inpatient and outpatient diagnosis and therapy. Long-term monitoring of ambulant or potentially ambulant inpatients provides significant potential for reducing the length of stay for post-coronary infarct patients in the intensive care setting and may result in earlier discharge from the hospital with greater assurance of safety to the patients. The indications for the use of this technique, noted below, are similar for both inpatients and outpatients.

The long-term EKG has proven effective in detecting transient episodes of cardiac dysrhythmia and in permitting the correlation of these episodes with cardiovascular symptomatology. It is also useful for patients who have symptoms of obscure etiology suggestive of cardiac arrhythmia. Examples of such symptoms include palpitations, chest pain, dizziness, light-headedness, near syncope, syncope, transient ischemic episodes, dyspnea, and shortness of breath.

This technique would also be appropriate at the time of institution of any arrhythmic drug therapy and may be performed during the course of therapy to evaluate response. It is also appropriate for evaluating a change of dosage and may be indicated shortly before and after the discontinuation of anti-arrhythmic medication. The therapeutic response to a drug whose duration of action and peak of effectiveness is defined in hours cannot be properly assessed by examining 30-40 cycles on a standard EKG rhythm strip. The knowledge that all patients placed on anti-arrhythmic medication do not respond to therapy and the known toxicity of anti-arrhythmic agents clearly indicate that proper assessment should be made on an individual basis to determine whether medication should be continued and at what dosage level.

The long-term EKG is also valuable in the assessment of patients with coronary artery disease. It enables the documentation of etiology of such symptoms as chest pain and shortness of breath. Since the standard EKG is often normal during the intervals between the episodes of precordial pain, it is essential to obtain EKG information while the symptoms are occurring. The long-term EKG has enabled the correlation of chest symptoms with the objective evidence of ST-segment abnormalities. It is appropriate for patients who are recovering from an acute myocardial infarction or coronary insufficiency before and after discharge from the hospital, since it is impossible to predict which of these patients is subject to ventricular arrhythmias on the basis of the presence or absence of rhythm disturbances during the period of initial coronary care. The long-term EKG enables the physician to identify patients who are at a higher risk of dying suddenly in the period following an acute myocardial infarction. It may also be reasonable and necessary where the high-risk patient with known cardiovascular disease advances to a substantially higher level

of activity which might trigger increased or new types of arrhythmias necessitating treatment. Such a high-risk case would be one in which there is documentation that acute phase arrhythmias have not totally disappeared during the period of convalescence.

The use of the long-term EKG for routine assessment of pacemaker function can no longer be justified, (see §20.8.1). Its use for the patient with an internal pacemaker would be covered only when he has symptoms suggestive of arrhythmia not revealed by the standard EKG or rhythm strip.

These guidelines are intended as a general outline of the circumstances under which the use of this diagnostic procedure would be warranted. Each patient receiving a long-term EKG should be evaluated completely, prior to performance of this diagnostic study. A complete history and physical examination should be obtained and the referring physician should review the indications for use of the long-term EKG.

The performance of a long-term EKG does not necessarily require the prior performance of a standard EKG. Nor does the demonstration of a normal standard EKG preclude the need for a long-term EKG. Finally, the demonstration of an abnormal standard EKG does not obviate the need for a long-term EKG if there is suspicion that the dysrhythmia is transient in nature.

A period of recording of up to 24 hours would normally be adequate to detect most transient arrhythmias and provide essential diagnostic information. The medical necessity for longer periods of monitoring must be documented.

Medical documentation for adjudicating claims for the use of the long-term EKG should be similar to other EKG services, X-ray services, and laboratory procedures. Generally, a statement of the diagnostic impression of the referring physician with an indication of the patient's relevant signs and symptoms should be sufficient for purposes of making a determination regarding the reasonableness and medical necessity for the use of this procedure. However, the intermediaries or carriers should require whatever additional documentation their medical consultants deem necessary to properly adjudicate the individual claim where the information submitted is not adequate.

It should be noted that the recording device furnished to the patient is simply one component of the diagnostic system and a separate charge for it will not be recognized under the durable medical equipment benefit.

Patient-Activated EKG Recorders, distributed under a variety of brand names, permit the patient to record an EKG upon manifestation of symptoms, or in response to a physician's order (e.g., immediately following strong exertion). Most such devices also permit the patient to simultaneously voice-record in order to describe symptoms and/or activity. In addition, some of these devices permit transtelephonic transmission of the recording to a physician's office, clinic, hospital, etc., having a decoder/recorder for review and analysis, thus

eliminating the need to physically transport the tape. Some of these devices also permit a "time sampling" mode of operation. However, the "time sampling" mode is not covered—only the patient-activated mode of operation, when used for the indications described below, is covered at this time.

Services in connection with patient-activated EKG recorders are covered when used as an alternative to the long-term EKG monitoring (described above) for similar indications - detecting and characterizing symptomatic arrhythmias, regulation of anti-arrhythmic drug therapy, etc. Like long-term EKG monitoring, use of these devices is covered for evaluating patients with symptoms of obscure etiology suggestive of cardiac arrhythmia such as palpitations, chest pain, dizziness, lightheadedness, near syncope, syncope, transient ischemic episodes, dyspnea and shortness of breath.

As with long-term EKG monitors, patient-activated EKG recorders may be useful for both inpatient and outpatient diagnosis and therapy. While useful for assessing some post-coronary infarct patients in the hospital setting, these devices should not, however, be covered for outpatient monitoring of recently discharged post-infarct patients.

Computer Analyzed Electrocardiograms - Computer interpretation of EKG's is recognized as a valid and effective technique which will improve the quality and availability of cardiology services. Reimbursement may be made for such computer service when furnished in the setting and under the circumstances required for coverage of other electrocardiographic services. Where either a laboratory's or a portable x-ray supplier's charge for EKG services includes the physician review and certification of the printout as well as the computer interpretation, the certifying physician must be identified on the Form CMS-1490 before the entire charge can be considered a reimbursable charge. Where the laboratory's (or portable x-ray supplier's) reviewing physician is not identified, the carrier should conclude that no professional component is involved and make its charge determination accordingly. If the supplying laboratory (or portable x-ray supplier when supplied by such a facility) does not include professional review and certification of the hard copy, a charge by the patient's physician may be recognized for the service. In any case the charge for the physician component should be substantially less than that for physician interpretation of the conventional EKG tracing in view of markedly reduced demand on the physician's time where computer interpretation is involved. Considering the unit cost reduction expected of this innovation, the total charge for the complete EKG service (taking of tracing and interpretation) when computer interpretation is employed should never exceed that considered reasonable for the service when physician interpretation is involved.

Transtelephonic Electrocardiographic Transmissions (Formerly Referred to as EKG Telephone Reporter Systems) coverage is extended to include the use of transtelephonic electrocardiographic (EKG) transmissions as a diagnostic service for the indications described below, when performed with equipment meeting the standards described below,

subject to the limitations and conditions specified below. Coverage is further limited to the amounts payable with respect to the physician's service in interpreting the results of such transmissions, including charges for rental of the equipment. The device used by the beneficiary is part of a total diagnostic system and is not considered durable medical equipment.

1 - Covered Uses

The use of transtelephonic EKGs is covered for the following uses:

- To detect, characterize, and document symptomatic transient arrhythmias;
- To overcome problems in regulating antiarrhythmic drug dosage;
- To carry out early posthospital monitoring of patients discharged after myocardial infarction; (only if 24-hour coverage is provided, see subsection 4, "Twenty-Four Hour Coverage," below).

Since cardiology is a rapidly changing field, some uses other than those specified above may be covered if, in the judgment of the contractor's medical consultants, such a use was justifiable in the particular case. The enumerated uses above represent uses for which a firm coverage determination has been made, and for which contractors may make payment without extensive claims development or review.

2 - Specifications for Devices

The devices used by the patient are highly portable (usually pocket-sized) and detect and convert the normal EKG signal so that it can be transmitted via ordinary telephone apparatus to a receiving station. At the receiving end, the signal is decoded and transcribed into a conventional EKG. There are numerous devices available which transmit EKG readings in this fashion. For purposes of Medicare coverage, however, the transmitting devices must meet at least the following criteria:

- They must be capable of transmitting EKG Leads, I, II, or III;
- These lead transmissions must be sufficiently comparable to readings obtained by a conventional EKG to permit proper interpretation of abnormal cardiac rhythms.

3 - Potential for Abuse - Need for Screening Guidelines

While the use of these devices may often compare favorably with more costly alternatives, this is the case only where the information they contribute is actively utilized by a knowledgeable practitioner as part of overall medical management of the patient. Consequently, it is vital that contractors be aware of the potential for abuse of these devices, and adopt necessary screening and physician education policies to detect and halt potentially abusive situations. For example, use of these devices to diagnose and treat suspected arrhythmias as a routine substitute for more conventional methods of diagnosis,

such as a careful history, physical examination, and standard EKG and rhythm strip would not be appropriate. Moreover, contractors should require written justification for use of such devices in excess of 30 consecutive days in cases involving detection of transient arrhythmias.

Contractors may find it useful to review claims for these devices with a view toward detecting patterns of practice which may be useful in developing schedules which may be adopted for screening such claims in the future.

4 - Twenty-Four Hour Coverage

No payment may be made for the use of these devices to carry out early posthospital monitoring of patients discharged after myocardial infarction unless provision is made for 24 hour coverage in the manner described below.

Twenty-four hour coverage means that there must be, at the monitoring site (or sites) an experienced EKG technician receiving calls; tape recording devices do not meet this requirement. Further, such technicians should have immediate access to a physician, and have been instructed in when and how to contact available facilities to assist the patient in case of emergencies.

Cross Reference

Medicare Benefit Policy Manual, Chapter 1, "Inpatient Hospital Services," §50. Medicare Benefit Policy Manual, Chapter 6, "Hospital Services Covered Under Part B," §10 and §20.3. Medicare Benefit Policy Manual, Chapter 15, "Covered Medical and Other Health Services," §§60.1 and 250. Medicare Claims Processing Manual, Chapter 16, "Laboratory Services From Independent Labs, Physicians, and Providers," §10.

National Coverage Analyses (NCAs)

This NCD has been or is currently being reviewed under the National Coverage Determination process. The following are existing associations with NCAs, from the National Coverage Analyses database.

- [Original Consideration for Electrocardiographic Services \(CAG-00158N\)](#) 

Additional Information

Other Versions

Title	Version	Effective Between	
Electrocardiographic Services	2	08/26/2004 - N/A	View
Electrocardiographic Services	1	06/01/1985 - 08/26/2004	You are here